

BG77xA-GL&BG95xA-GL

GNSS Application Note

LPWA Module Series

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About the Document

Revision History

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Contents

About the Document.....	3
Contents.....	4
Table Index.....	7
Figure Index.....	8
1 Introduction	9
1.1. Applicable Modules.....	9
1.2. Supported NMEA Sentence Types	10
1.3. GNSS and WWAN Coexistence Management.....	11
1.3.1. WWAN Priority Mode	11
1.3.2. GNSS Priority Mode.....	13
1.3.3. WWAN and GNSS Switching Delays.....	14
1.3.4. GNSS/WWAN Priority and PSM	15
1.3.5. GNSS/WWAN Priority and eDRX	16
1.3.6. Minimum eDRX Cycle Recommendation in WWAN Priority Mode	17
1.4. XTRA Feature	18
1.4.1. XTRA Configuration Items	18
1.4.2. XTRA Time Source.....	19
1.4.3. Downloading XTRA File	20
1.4.3.1. Initiate XTRA File Downloading	20
1.4.3.2. Check XTRA File Validity	20
1.4.3.3. Download XTRA File	21
1.4.4. Update XTRA File	23
1.4.4.1. XTRA File Downloading Triggered with AT+QGPSCFG="xtra_download",1	23
1.4.4.2. Automatic XTRA File Downloading by Launching GNSS.....	23
1.4.4.3. CEP Proxy Server Connection.....	23
2 Description of GNSS AT Commands	25
2.1. AT Command Introduction	25
2.1.1. Definitions.....	25
2.1.2. AT Command Syntax	25
2.2. Declaration of AT Command Examples	26
2.3. AT Command Description	26
2.3.1. AT+QGPSCFG Configure GNSS	26
2.3.1.1. AT+QGPSCFG="outport" Configure NMEA Output Port	27
2.3.1.2. AT+QGPSCFG="gnssconfig" Configure Supported GNSS Constellations	28
2.3.1.3. AT+QGPSCFG="nmeafmt" Configure NMEA Protocol Standard	29
2.3.1.4. AT+QGPSCFG="gpsnmeatype" Configure GPS NMEA Sentence Output	30
2.3.1.5. AT+QGPSCFG="glonassnmeatype" Configure GLONASS NMEA Sentence Output Type.....	31
2.3.1.6. AT+QGPSCFG="nmeasrc" Enable/Disable Acquisition of NMEA Sentences via AT+QGPSSGNMEA	32

2.3.1.7.	AT+QGPSCFG="autogps"	Enable/Disable Automatic Running of GNSS	32
2.3.1.8.	AT+QGPSCFG="priority"	Set GNSS or WWAN Priority Mode	33
2.3.1.9.	AT+QGPSCFG="xtrafilesize"	Select XTRA File to be Downloaded	34
2.3.1.10.	AT+QGPSCFG="xtra_info"	Query Existing XTRA File Information	35
2.3.1.11.	AT+QGPSCFG="gpsdop"	Query GNSS DOP Value	36
2.3.1.12.	AT+QGPSCFG="nmea_epe"	Enable/Disable EPE NMEA Sentence Outputting	37
2.3.1.13.	AT+QGPSCFG="estimation_error"	Get Estimation Error	38
2.3.1.14.	AT+QGPSCFG="xtra_download"	Trigger XTRA File Downloading	38
2.3.1.15.	AT+QGPSCFG="xtra_autodownload"	Enable/Disable Automatic Downloading of XTRA File	39
2.3.1.16.	AT+QGPSCFG="xtra_cfg"	Configure XTRA Proxy Server	40
2.3.1.17.	AT+QGPSCFG="test_mode"	Enable/Disable XTRA Test Mode	41
2.3.1.18.	AT+QGPSCFG="gnss_mode"	Select GNSS Mode	41
2.3.2.	AT+QGPS	Turn on GNSS	42
2.3.3.	AT+QGPSEND	Turn off GNSS	44
2.3.4.	AT+QGPSLOC	Acquire Positioning Information	45
2.3.5.	AT+QGPSGNMEA	Acquire NMEA Sentences	47
2.3.6.	AT+QCFGEXT	Query and Configure Extended Settings	49
2.3.6.1.	AT+QCFGEXT="addgeo"	Add a Geo-fence	49
2.3.6.2.	AT+QCFGEXT="deletegeo"	Delete a Geo-fence	52
2.3.6.3.	AT+QCFGEXT="querygeo"	Query Position Relative to Geo-Fence	52
2.3.7.	AT+QGPSXTRA	Enable/Disable XTRA Function	53
2.3.8.	AT+QGPSXTRATIME	Inject XTRA Time	54
2.3.9.	AT+QGPSXTRADATA	Inject Assistance Data to GNSS Engine	55
2.3.10.	AT+QGPSDEL	Delete Assistance Data	56
2.4.	BG951A-GL-Specific AT Command Description		57
2.4.1.	AT+QGPSADATA	Query and Configure LLE Data	57
2.4.1.1.	AT+QGPSADATA="erase"	Erase LLE Data	58
2.4.1.2.	AT+QGPSADATA="write"	Write LLE Data	58
2.4.2.	AT+QGPSRM	Enable/Disable RF Mute	59
2.4.3.	AT+QGPSPPS	Enable/Disable 1PPS Output	60
2.4.4.	AT+QGPSRST	Reset GNSS	60
2.4.5.	AT+QGPSPSM	Sleep and Wake up GNSS	61
3	Examples		63
3.1.	Turn on/off GNSS		63
3.2.	Acquire Positioning Information		63
3.3.	Query Satellite System		65
3.4.	Implement Geo-Fence Function		65
3.5.	How to Use XTRA Feature		66
3.5.1.	XTRA File Download is Triggered with AT+QGPSCFG="xtra_download",1		66
3.5.2.	XTRA File Downloading Triggered by Starting GNSS		67
4	Use Cases		68

4.1.	Enable PSM	68
4.2.	Recommended Mode: WWAN Priority Mode	68
4.3.	Recommended Mode: GNSS Priority Mode	69
4.4.	GNSS Limitations	69
5	FAQ	71
6	Summary of Error Codes	72
7	Appendix References	74

Table Index

Table 1: Applicable Modules.....	9
Table 2: Average Delays Captured for Different RATs (Unit: s)	14
Table 3: Recommended Minimum eDRX Cycle in WWAN Priority Mode (Unit: s).....	18
Table 4: CEP File Size (BG77xA-GL and BG95xA-GL Modules Except BG951A-GL)	19
Table 5: Types of AT Commands	25
Table 6: Summary of Error Codes.....	72
Table 7: Related Documents	74
Table 8: Terms and Abbreviations	74

Figure Index

Figure 1: GNSS and WWAN Coexistence Management (WWAN Priority Mode)	12
Figure 2: GNSS and WWAN Coexistence Management (GNSS Priority Mode)	13
Figure 3: Configure PSM in GNSS Priority Mode	16
Figure 4: Configure eDRX in GNSS Priority Mode	17
Figure 5: eDRX Diagram.....	18
Figure 6: XTRA File Downloading Process.....	22

1 Introduction

Quectel BG77xA-GL and BG95xA-GL modules include a GNSS engine, with the XTRA (i.e., gpsOneXTRA) feature. The GNSS engine is suitable for various applications that require accurate positioning at the lowest cost, or position tracking without network assistance.

Except BG951A-GL, all other BG77xA-GL and BG95xA-GL modules feature a fully integrated GNSS engine that supports GPS and GLONASS systems. The GNSS engine can receive simultaneous signals from GPS and GLONASS at any given time. These modules are based on a cost-optimized architecture in which WWAN (Cat M1 and Cat NB2) and GNSS Rx chains share certain hardware blocks. However, these modules do not support concurrent operation of WWAN and GNSS.

BG951A-GL features an independent GNSS engine that supports GPS, GLONASS, Galileo, BDS and QZSS systems. The GNSS engine can receive simultaneous signals from maximum two constellations (GPS and another constellation) at any given time. This module supports concurrent operation of WWAN and GNSS, which means that position tracking with network assistance is also possible.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Model
BG77xA-GL	BG770A-GL
	BG773A-GL
BG95xA-GL	BG950A-GL
	BG951A-GL
	BG953A-GL
	BG955A-GL

1.2. Supported NMEA Sentence Types

NMEA sentences supported by default are compliant with the NMEA 0183 version 4.10 standard. As illustrated below, each satellite system uses distinct prefixes, so that NMEA sentences of different satellite systems can be easily distinguished.

GPS sentences have the prefix "GP":

- GPGBA – Global positioning system fix data, such as time and position.
- GPRMC – Recommended minimum specific GNSS data.
- GPGSV – GNSS satellites in view (number of satellites in view, satellite ID numbers, etc.).
- GPGSA – GNSS DOP and active satellites.
- GPVTG – Course over ground and ground speed.

GLONASS sentences have the prefixes "GL" and "GN" ("GN" is a general NMEA statement when GNSS is working with more than one constellation):

- GLGSV – GNSS satellites in view (number of satellites in view, satellite ID numbers, etc.).
- GNGSA – GNSS DOP and active satellites.

For Galileo sentences, the prefixes are "GA" and "GN":

- GAGSV – GNSS satellites in view (number of satellites in view, satellite ID numbers, etc.).
- GNGSA – GNSS DOP and active satellites

For BDS sentences, the prefixes are "BD" and "GN":

- BDGSV – GNSS satellites in view (number of satellites in view, satellite ID numbers, etc.).
- GNGSA – GNSS DOP and active satellites

For QZSS sentences, the prefixes are "GQ" and "GN":

- GQGSV - GNSS satellites in view (number of satellites in view, satellite ID numbers, etc.).
- GNGSA – GNSS DOP and active satellites

NOTE

Galileo, BDS and QZSS are supported on BG951A-GL only. By default, both GSV and GSA sentences are outputted for Galileo, BDS and QZSS.

1.3. GNSS and WWAN Coexistence Management

As GNSS and WWAN cannot work simultaneously on BG77xA-GL and BG95xA-GL modules, except on BG951A-GL, either WWAN or GNSS takes priority during implementation. By default, the modules are configured into the GNSS priority mode, which can be switched to the WWAN priority mode with **AT+QGPSCFG="priority"**. The command takes effect immediately. For more information, see **Chapter 2.3.1.8**.

This chapter does not apply to BG951A-GL since it supports simultaneous operation of GNSS and WWAN.

1.3.1. WWAN Priority Mode

In WWAN priority mode, GNSS positioning request succeeds only after RRC is released and after WWAN has finished transferring all data and enters idle state. Features of WWAN priority mode:

- Data service works well all the time (No WWAN page is missed).
- RRC connection is not impacted by GNSS operation.
- GNSS session is deferred to the time when the UE goes to eDRX state ¹⁾.

For LPWA applications, the duration for WWAN active status is fairly short. Therefore, GNSS has an opportunity to get a position fix.

NOTE

1. ¹⁾ The configured eDRX cycle must be sufficiently long. In WWAN priority mode, if eDRX Sleep Cycle is shorter than $t + t_1$, GNSS will not be able to get a position fix. For more information, see **Chapter 1.3.6**.
2. When eDRX Sleep Cycle is less than $t + t_1$ or eDRX is not supported, it is recommended to set the module into GNSS priority mode.
3. When switching from the WWAN priority mode to the GNSS priority mode, it takes some time (under the open sky, see **Chapter 1.3.3**) for GNSS to get a position fix (if GNSS has got a position fix before).

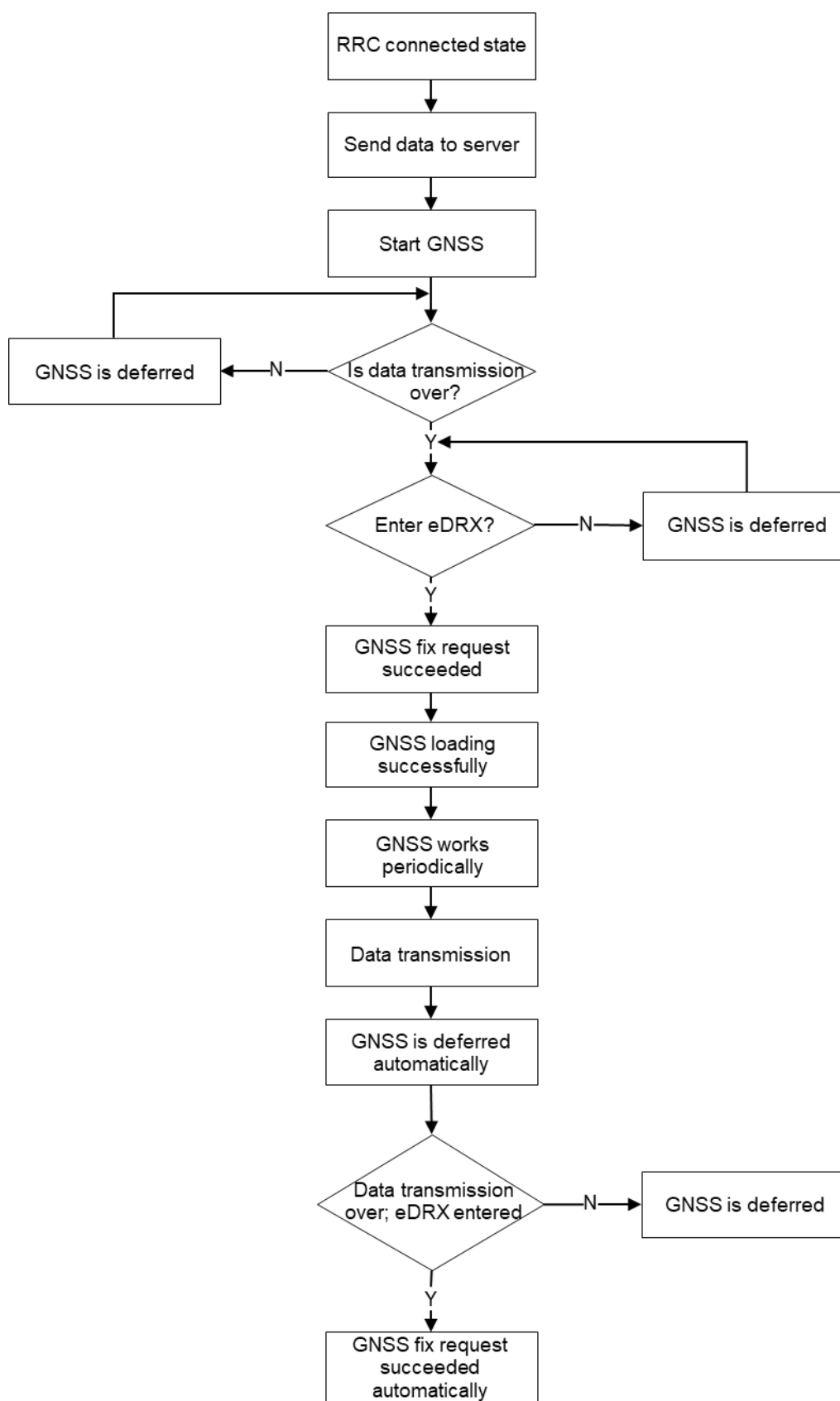


Figure 1: GNSS and WWAN Coexistence Management (WWAN Priority Mode)

1.3.2. GNSS Priority Mode

In GNSS priority mode, GNSS positioning request succeeds only in WWAN unloaded states:

- **UE in RRC connected state:**

UE locally releases RRC connection, unloads WWAN, and initiates a GNSS session. After the GNSS session is completed, WWAN will be loaded and the RRC connection will be initiated again. If a data transmission was interrupted by the GNSS session, after RRC reinitiates, the data transmission will be restarted.

- **UE in eDRX state:**

Because of WWAN unloaded state, UE will miss pages sent on the paging channel while the GNSS session is still in progress.

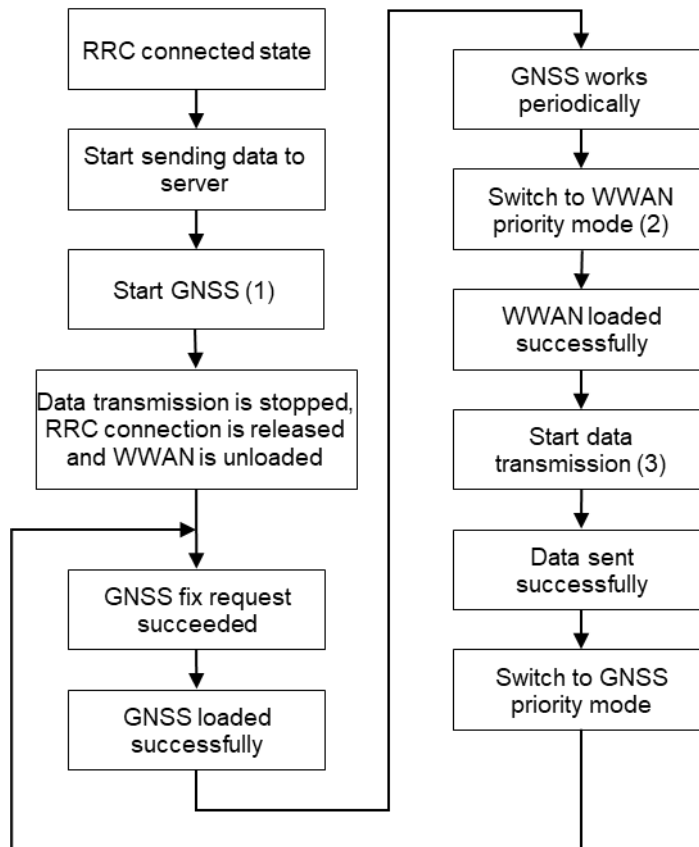


Figure 2: GNSS and WWAN Coexistence Management (GNSS Priority Mode)

NOTE

1. GNSS starts working as soon as it is launched.
2. After switching to WWAN priority mode, GNSS is stopped automatically. In such a case, the power consumption is the same as when GNSS is stopped with **AT+QGPSEND**.
3. It is recommended to delay data transmission for some time, since the module needs this time to switch from GNSS to WWAN priority mode (see **Chapter 1.3.3**).
4. The following operations/procedures have higher priority than GNSS/WWAN:
 - Power-off/Power Saving Mode
 - PS detaching

1.3.3. WWAN and GNSS Switching Delays

The following table summarizes the average delays captured for different RATs. All tests were executed by transmitting 500-byte data under open sky. The presented average delays are given for illustrative purposes only and should be taken with reservations.

● Test Environment

GNSS CNR = TBD

LTE-M RSRP = TBD

SINR = TBD

NB-IoT RSRP = TBD

SINR = TBD

● Test Steps

Load WWAN → Unload WWAN → Load GNSS → Get a position fix → Unload GNSS → Load WWAN → Transfer WWAN data.

Table 2: Average Delays Captured for Different RATs (Unit: s)

Network Type	LTE-M	NB-IoT	GSM (BG955A-GL Only)
Load WWAN	TBD	TBD	TBD
Unload WWAN	TBD	TBD	TBD
Load GNSS	TBD	TBD	TBD
Unload GNSS	TBD	TBD	TBD
Delay Between "Switching to WWAN Priority Mode" and "Start Sending Data" ¹⁾	TBD	TBD	TBD

Time to Transfer WWAN Data (delay for setting up and closing connection is not considered) ²⁾	TBD	TBD	TBD
Delay Between "Switching to GNSS Priority Mode" and "Getting GNSS Started"	TBD	TBD	TBD
GNSS Fix with/without XTRA	TBD	TBD	TBD

NOTE

1. "Load" and "Unload" mean loading/unloading of WWAN/GNSS protocol stack.
2. ¹⁾ Once WWAN is loaded successfully (you can query it via **AT+QGPSCFG="priority"**), the data can be sent.
3. ²⁾ The time for transferring WWAN data varies depending on the coverage level and data amount. In this example, 500-byte data was transferred.

1.3.4. GNSS/WWAN Priority and PSM

In GNSS priority mode, PSM cannot be configured when GNSS is active. However, it can be configured before starting GNSS. After PSM is configured successfully, the modules can enter PSM even when the GNSS is active.

In WWAN priority mode, the modules can configure and enter PSM regardless of whether GNSS is active or not.

For the actual scenario with PSM, see **Chapter 4.1**.

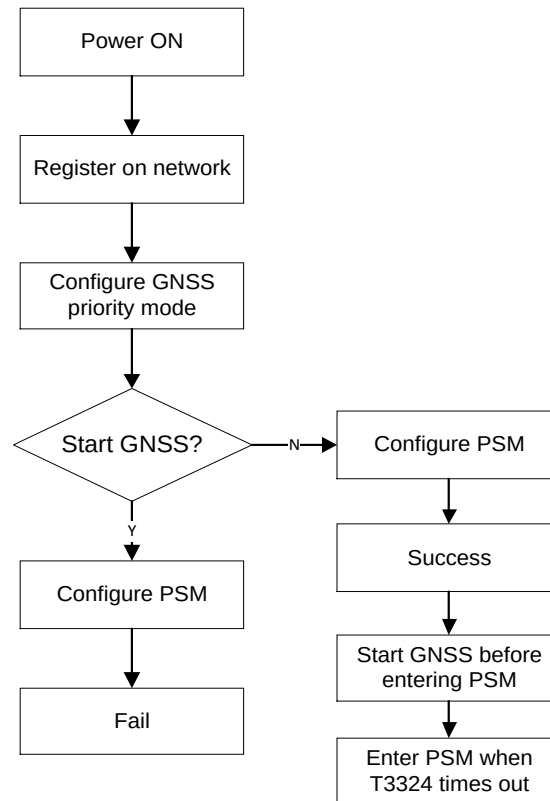


Figure 3: Configure PSM in GNSS Priority Mode

1.3.5. GNSS/WWAN Priority and eDRX

eDRX can be configured in GNSS priority mode when GNSS is turned off. However, eDRX cannot be configured in GNSS priority mode when GNSS is active. In this case, the modules cannot enter eDRX even if it has been configured successfully before starting GNSS.

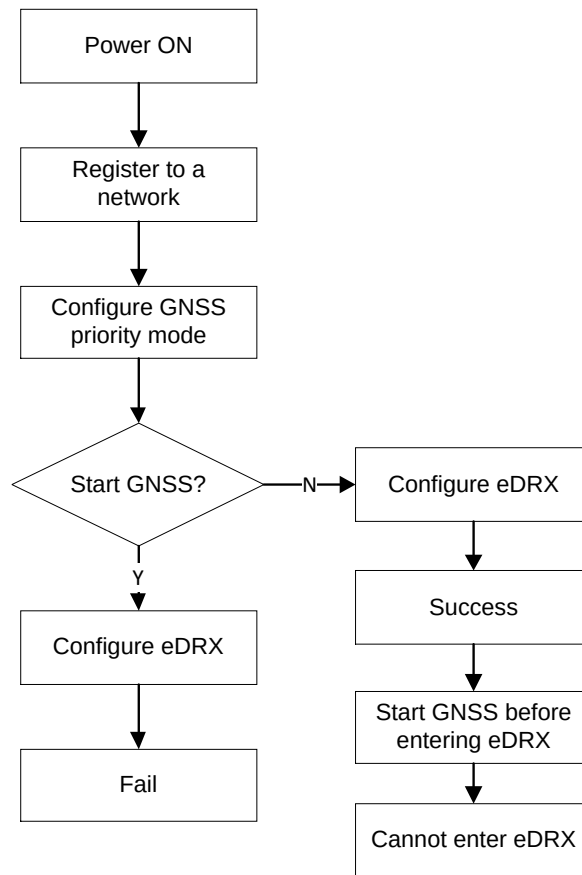


Figure 4: Configure eDRX in GNSS Priority Mode

In WWAN priority mode, the modules are able to configure and enter eDRX regardless of whether GNSS is active or not.

1.3.6. Minimum eDRX Cycle Recommendation in WWAN Priority Mode

Assuming that a GNSS fix on average takes maximum t seconds:

$$t1 = \text{GNSS Unloading Time} + \text{WWAN Loading Time} + \text{Page Accepting Wake-up Time}$$

From test observations, $t1 \approx 1.5$ seconds.

The recommended minimum eDRX Sleep Cycle should be $> (t + t1)$ seconds.

$$\text{eDRX Sleep Cycle} = \text{eDRX Cycle} - \text{PTW}$$

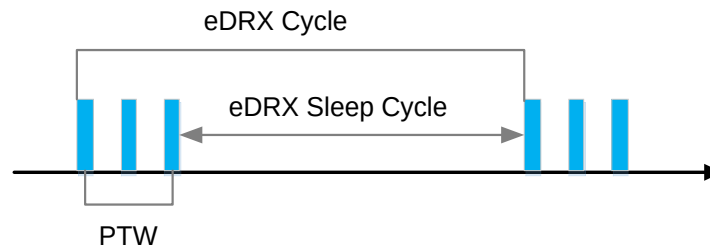


Figure 5: eDRX Diagram

Table 3: Recommended Minimum eDRX Cycle in WWAN Priority Mode (Unit: s)

Network Type	LTE-M	NB-IoT	GSM (For BG955A-GL Only)
Common eDRX Cycles	TBD	TBD	TBD
Recommended Minimum eDRX Cycle (Cold start with XTRA)	TBD	TBD	TBD
Recommended Minimum eDRX Cycle (Cold start without XTRA)	TBD	TBD	TBD

NOTE

The recommended minimum eDRX cycle is based on strong signal conditions. It is necessary to extend eDRX Sleep Cycle in weak signal conditions.

1.4. XTRA Feature

XTRA technology enhances GNSS performance and provides simplified GNSS assistance delivery, including ephemeris, almanac, ionosphere, UTC, health, and coarse time assistance for GNSS engine. After activating XTRA, the TTFF (Time to First Fix) can be reduced under open sky.

- For BG77xA-GL and BG95xA-GL modules, except BG951A-GL, XTRA feature implies CEP file downloading.
- For BG951A-GL module, XTRA feature implies LLE file downloading for chosen constellations.

1.4.1. XTRA Configuration Items

Select the XTRA file to be downloaded via **AT+QGPSCFG="xtrafilesize"**. XTRA file size and type are shown in the following tables.

Table 4: CEP File Size (BG77xA-GL and BG95xA-GL Modules Except BG951A-GL)

File Type & Validity	7-Day File (kB)	3-Day File (kB)	2-Day File (kB)	1-Day File (kB)
GPS + GLONASS	57.3	24.5	18.4	10.2

Table 4: LLE File Size (BG951A-GL)

File Type & Validity	14-Day File (kB)	7-Day File (kB)	3-Day File (kB)
GPS	36.8	23.7	16.2
GLONASS	25.5	16.4	11.2
BDS	38.2	24.6	16.8
Galileo	26.9	17.3	11.8
QZSS	1.4	0.9	0.6

NOTE

The sizes listed in the tables above are for illustrative purposes only.

1.4.2. XTRA Time Source

XTRA time format is UTC time, which can be obtained by:

- Injecting time manually
- Synchronizing NITZ time
- Synchronizing GPS time
- Synchronizing NTP time

XTRA time is used to check XTRA file validity and inject ephemeris data at the current time to the GNSS engine. Unless you inject the time manually, the XTRA time will be injected automatically before injecting XTRA data. The module first starts synchronizing NITZ time. If NITZ time is not available, the module will synchronize GPS time. Lastly, if neither NITZ nor GPS time is available, the module will try to obtain NTP time.

NOTE

time.xtracloud.net is the NTP time server. When obtaining NTP time, the module starts a 35-second timer. If NTP time is not obtained within 35 seconds, the XTRA technology will not be available and the module will launch GNSS without XTRA.

1.4.3. Downloading XTRA File

1.4.3.1. Initiate XTRA File Downloading

XTRA files are located on XTRA servers that are not configurable. To download an XTRA file, the data service and the XTRA feature have to be enabled.

There are two ways to trigger XTRA file downloading:

- Execute **AT+QGPSCFG="xtra_download",1**
- Start GNSS ¹⁾ with **AT+QGPS=1**

When XTRA file downloading is triggered, the module starts a 35-second timer. If no XTRA file is downloaded within 35 seconds, the XTRA technology will not be available and the module will launch GNSS without XTRA.

NOTE

1. ¹⁾ XTRA file downloading is triggered automatically when starting GNSS with **AT+QGPS=1**.
2. If the XTRA file is downloaded via **AT+QGPSCFG="xtra_download",1** successfully, starting GNSS with **AT+QGPS=1** will not trigger downloading of the XTRA file again. If no valid XTRA file is available, starting GNSS with **AT+QGPS=1** triggers automatic XTRA file downloading.
3. If the network is unavailable when starting GNSS, the module skips downloading XTRA file and starts GNSS without XTRA.
4. The module checks XTRA file validity before downloading. If the existing XTRA file has expired, the module downloads the new XTRA file and then reports the URC **+QGPSURC:"XTRA_DL",<DL_err>** indicating the result of the XTRA file download.

<DL_err> Integer type. Downloading result.
 0 XTRA file downloaded successfully.
 -1 Downloading failed.

1.4.3.2. Check XTRA File Validity

When XTRA file downloading is triggered, the module checks existing XTRA file validity. If the existing XTRA file is valid, the module stops downloading the new XTRA file. You can check existing XTRA file

validity manually by executing **AT+QGPSCFG="xtra_info"** (see **Chapter 2.3.1.10**) or **AT+QGPSXTRADATA?** (see **Chapter 0**).

1.4.3.3. Download XTRA File

After XTRA file downloading is triggered automatically by starting GNSS with **AT+QGPS=1**, or manually by executing **AT+QGPSCFG="xtra_download",1**, the module downloads the XTRA file with a 3-day validity ¹⁾ by default and then injects the downloaded XTRA file into the GNSS engine. XTRA file downloading process is illustrated in the figure below:

NOTE

¹⁾ You can change XTRA file validity with **AT+QGPSCFG="xtrafilesize"**. For more information, see **Chapter 2.3.1.9**.

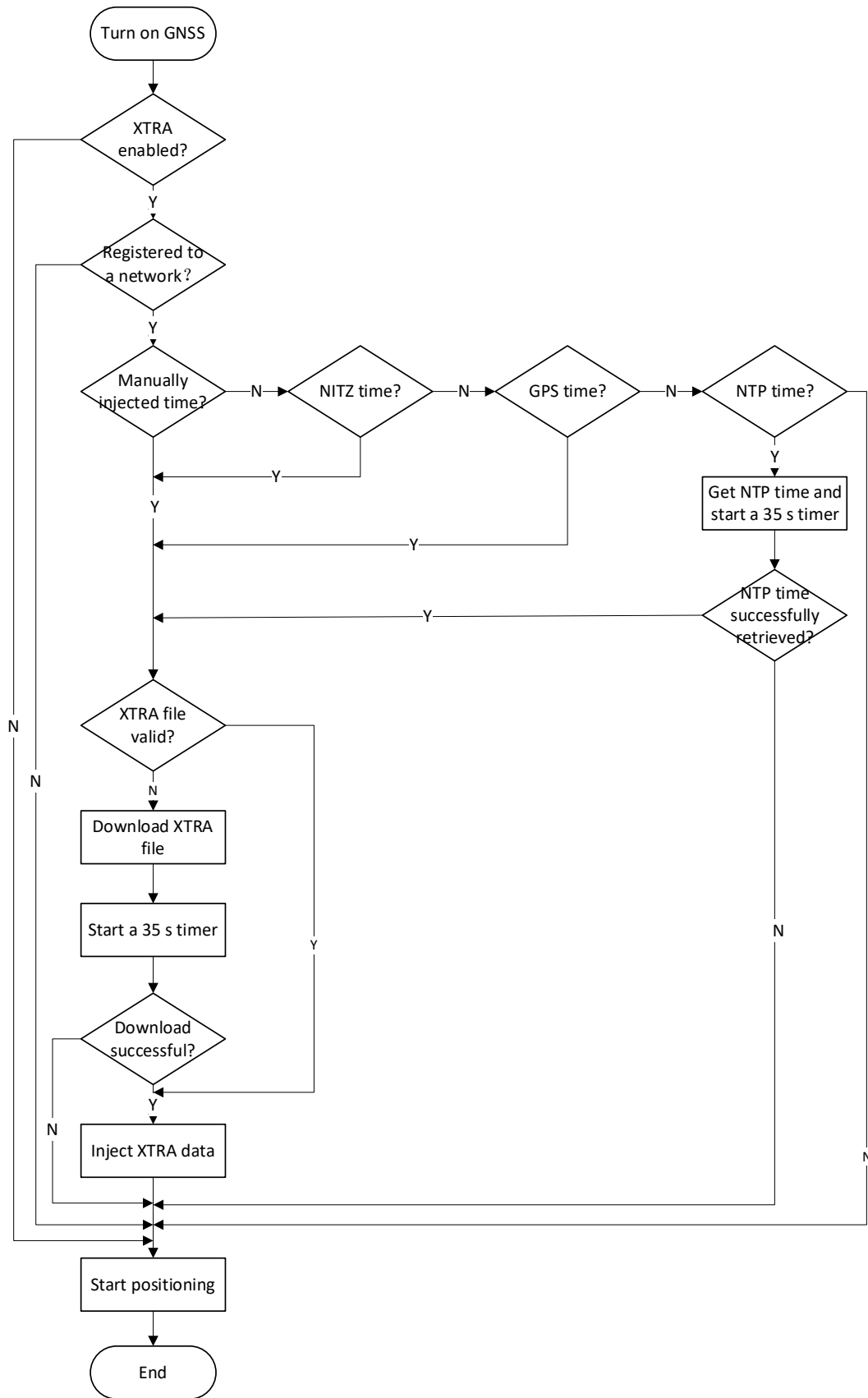


Figure 6: XTRA File Downloading Process

1.4.4. Update XTRA File

XTRA file must be updated regularly. If GNSS is started by executing **AT+QGPS=1** or **AT+QGPSCFG="xtra_download",1**, the module checks existing XTRA file validity. If the existing XTRA file has expired, the new XTRA file is downloaded and injected automatically.

1.4.4.1. XTRA File Downloading Triggered with AT+QGPSCFG="xtra_download",1

- Step 1:** Enable the XTRA feature with **AT+QGPSXTRA=1** and then restart the module to activate the feature.
- Step 2:** Check the network registration status with **AT+CEREG?** (see *document [1]* for detailed information) before triggering XTRA file downloading.
- Step 3:** Check XTRA time with **AT+QGPSXTRATIME?**. If there is no XTRA time, inject XTRA time manually.
- Step 4:** Execute **AT+QGPSCFG="xtra_download",1** to trigger XTRA file downloading.
- Step 5:** Start GNSS with **AT+QGPS=1**.

1.4.4.2. Automatic XTRA File Downloading by Launching GNSS

- Step 1:** Enable the XTRA feature with **AT+QGPSXTRA=1** and then restart the module to activate the feature.
- Step 2:** Check the network registration status with **AT+CEREG?** before triggering XTRA file downloading.
- Step 3:** Check XTRA time with **AT+QGPSXTRATIME?**. If there is no XTRA time, inject XTRA time manually.
- Step 4:** Start GNSS with **AT+QGPS=1** to trigger automatic XTRA file downloading.

1.4.4.3. CEP Proxy Server Connection

CEP (Connected Ephemeris Prediction) File

The CEP file contains critical satellite GPS ephemeris information for Sony's GNSS receivers. Obtaining the CEP files involves accessing a Sony CEP server. To receive the necessary access credentials for the Sony CEP server, please contact Quectel Technical Support. Note: At this point, assisted ephemeris information is supported for GPS constellation only.

Architecture for Device OEM

It is recommended for the device OEM to establish a proxy server capable of pushing and distributing the CEP file to the OEM edge devices. The proxy server should retrieve and update the CEP file from the Sony server and cache it based on the validity duration specified for each file. The integration of a proxy server to manage and distribute CEP files is a recommended approach to ensure efficient retrieval and utilization of the necessary satellite data to the edge OEM devices.

Access to Sony Server

Access to the Sony server and downloading the CEP file should be done by HTTP protocol. Account (username/password) should be obtained from your Sony customer representative, free of charge.

- By browser: <https://asp.gigacc.com/user/gps-cep>
- By wget: [https://asp.gigacc.com/webdav/top/1513^cep_pak_1 day/cep_pak.bin](https://asp.gigacc.com/webdav/top/1513^cep_pak_1%20day/cep_pak.bin)

When acquiring CEP file by wget, the username must be set as below:

Username: xxxxxxxxxx gps_cep ("gps_cep" must follow the user name)

Password: xxxxxxxxxx

CEP Service Evaluation

Sony chips allow potential users of CEP service an easier way to test it before signing the agreement with Sony and assembling the proxy-server topology. Integrated in Sony are pre-approved username and password which allow access to CEP server via HTTP protocol. With AT commands (or CAPIs from Sony iMCU) CEP files in various durations are available for download for evaluation of this service. Since communication with Sony server is established per each request and not mitigated by proxy-server, excessive usage of these commands may lead to suspension and even cancellation of these credentials. To prevent such incidents, customers using these credentials should not use them as a part of testing automation or to test the availability and stability of Sony server. Once it is decided to use CEP service, Sony customer representative should be contacted to retrieve a unique username and password.

Available CEP File on Sony Server

CEP file can be downloaded via HTTP protocol. Files in Sony CEP server are updated once per day at 00:00 UTC (1-day CEP file is updated every 6 hours). It is recommended to access Sony server at 00:30 UTC onwards, to make sure file has already been updated.

2 Description of GNSS AT Commands

2.1. AT Command Introduction

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 5: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or list of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

2.3. AT Command Description

2.3.1. AT+QGPSCFG Configure GNSS

This command queries and configures various GNSS settings, including the NMEA output port and output types of NMEA sentences.

AT+QGPSCFG Configure GNSS	
Test Command AT+QGPSCFG=?	Response: +QGPSCFG: "outport",(list of supported <outport>s),(list of supported <baudrate>s) +QGPSCFG: "gnssconfig",(list of supported <GNSS_config>s) +QGPSCFG: "nmeafmt",(list of supported <NMEA_fmt_config>s) +QGPSCFG: "gpsnmeatype",(list of supported <GPS_NMEA_type>s) +QGPSCFG: "glonassnmeatype",(list of supported <GLONASS_NMEA_type>s) +QGPSCFG: "nmeasrc",(list of supported <NMEA_src>s) +QGPSCFG: "autogps",(list of supported <autoGPS>s) +QGPSCFG: "priority",(list of supported <priority_type>s),(list of supported <save>s) +QGPSCFG: "extrafilesize",(list of supported <type>s) +QGPSCFG: "extra_info" +QGPSCFG: "gpsdop" +QGPSCFG: "nmea_epe",(list of supported <NMEA_EPE>s) +QGPSCFG: "estimation_error" +QGPSCFG: "extra_download",(list of supported <type>s) +QGPSCFG: "extra_autodownload",(list of supported <mode>s) +QGPSCFG: "extra_cfg"(list of supported <URL>s),(list of supported <username>s),(list of supported <password>s) +QGPSCFG: "agnssjamming",,(list of supported <mode>s),(list of supported <cmp_counts>s),(list of supported <cmp_value>s) +QGPSCFG: "agnssjammingurcmode",(list of supported <URC_

	fmt>s) +QGPSCFG: "test_mode",(list of supported <mode>s) +QGPSCFG: "gnss_mode",(list of supported <mode>s) OK
Characteristics	-

NOTE

1. GNSS jamming feature and "agnssjamming" and "agnssjammingurcmode" are currently supported only on BG770A-GL and BG950A-GL. For more details see [document \[2\]](#).
2. "gnss_mode" is used to disable GNSS or configure GNSS mode to I2C or UART. It is supported only on BG951A-GL.

2.3.1.1. AT+QGPSCFG="output" Configure NMEA Output Port

This command queries and configures the NMEA output port and the port baud rate.

AT+QGPSCFG="output" Configure NMEA Output Port	
Write Command AT+QGPSCFG="output",[<output>],[<baudrate>]]	Response If the optional parameters are omitted, query the current setting. +QGPSCFG: "output",<output>,[<baudrate>] OK If any of the optional parameters is specified, set the NMEA output port and the port baud rate when <output> is "uartnmea" or "auxnmea". OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots. The configuration is saved automatically.

Parameter

<output>	String type. NMEA output port.
"none"	Close NMEA sentence output
"usbntmea"	Output via USB NMEA port

	"uartnmea"	Output via GNSS UART port (Supported only on BG951A-GL)
	"auxnmea"	Output via debug UART port on BG77xA-GL, or CLI UART port on BG95xA-GL
<baudrate>	Integer type. Baud rate of GNSS UART, debug UART and CLI UART ports. Unit: bps.	
	4800	
	9600	
	19200	
	38400	
	57600	
	<u>115200</u>	
	230400	
	460800	
	921600	
<errcode>	Integer type. Error code. See Chapter 6 for more information.	

NOTES

1. The **<baud_rate>** is available only when **<output>** is "uartnmea" or "auxnmea".
2. When **<baud_rate>** is 4800 or 9600, data loss may occur if a large amount of NMEA sentences is output.

2.3.1.2. AT+QGPSCFG="gnssconfig" Configure Supported GNSS Constellations

This command queries and configures the GNSS constellations supported by the module.

AT+QGPSCFG="gnssconfig" Configure Supported GNSS Constellations	
Write Command AT+QGPSCFG="gnssconfig"[,<GNSS_config>]	Response If the optional parameter is omitted, query the current setting: +QGPSCFG: "gnssconfig",<GNSS_config> OK If the optional parameter is specified, set the supported GNSS constellations: OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots. The configuration is saved automatically.

Parameter

<GNSS_config>	Integer type. Supported GNSS constellation. <u>1</u> GPS + GLONASS 2 GPS + BDS 3 GPS + Galileo 4 GPS + QZSS
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

On BG77xA-GL and BG95xA-GL modules except BG951A-GL, only GPS + GLONASS is supported.

2.3.1.3. AT+QGPSCFG="nmeafmt" Configure NMEA Protocol Standard

This command queries and configures the protocol standard of NMEA sentences.

AT+QGPSCFG="nmeafmt" Configure NMEA Protocol Standard	
Write Command AT+QGPSCFG="nmeafmt"[,<NMEA_fmt_config>]	Response If the optional parameter is omitted, query the current setting: +QGPSCFG: "nmeafmt",<NMEA_fmt_config> OK If the optional parameter is specified, set the NMEA protocol standard: OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<NMEA_fmt_config>	Integer type. NMEA protocol standards. <u>1</u> NMEA output sentences conform to NMEA 0183 version 4.10.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

2.3.1.4. AT+QGPSCFG="gpsnmeatype" Configure GPS NMEA Sentence Output

This command queries and configures the output type of GPS NMEA sentences.

AT+QGPSCFG="gpsnmeatype" Configure GPS NMEA Sentence Output Type	
Write Command AT+QGPSCFG="gpsnmeatype"[,<GPS_NMEA_type>]	Response If the optional parameter is omitted, query the current setting: +QGPSCFG: "gpsnmeatype",<GPS_NMEA_type> OK If the optional parameter is specified, set the output type of GPS NMEA sentences. OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<GPS_NMEA_type>	Integer type. Output type of GPS NMEA sentences by ORed. Range: 0–31.
0	Disable
1	GGA
2	RMC
4	GSV
8	GSA
16	VTG
31	All five types above
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

On BG951A-GL, when **<GPS_NMEA_type>=0**, all types of NMEA sentences including BDS, Galileo and QZSS will be disabled.

2.3.1.5. AT+QGPSCFG="glonassnmeatype" Configure GLONASS NMEA Sentence Output Type

This command queries and configures the output type of GLONASS NMEA sentences.

AT+QGPSCFG="glonassnmeatype" Configure GLONASS NMEA Sentence Output Type	
Write Command AT+QGPSCFG="glonassnmeatype"[, <GLONASS_NMEA_type>]	Response If the optional parameter is omitted, query the current setting: +QGPSCFG: "glonassnmeatype",<GLONASS_NMEA_type> OK If the optional parameter is specified, set the output type of GLONASS NMEA sentences. OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<GLONASS_NMEA_type>	Integer type. Output type of GLONASS NMEA sentences by ORed. Range: 0–3. 0 Disable 1 GSV 2 GSA 3 GSV + GSA
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

This command is not supported on BG951A-GL module.

2.3.1.6. AT+QGPSCFG="nmeasrc" Enable/Disable Acquisition of NMEA Sentences via

AT+QGPGNMEA

This command enables/disables the acquisition of NMEA sentences via **AT+QGPGNMEA** and queries the current settings.

AT+QGPSCFG="nmeasrc" Enable/Disable Acquisition of NMEA Sentences via AT+QGPGNMEA

Write Command AT+QGPSCFG="nmeasrc"[,<NMEA_src>]	Response If the optional parameter is omitted, query the current setting: +QGPSCFG: "nmeasrc",<NMEA_src> OK If the optional parameter is specified, enable/disable the acquisition of NMEA sentences via AT+QGPGNMEA . OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<NMEA_src>	Integer type. Sets whether to acquire the original NMEA sentences via AT+QGPGNMEA . 0 Disable <u>1</u> Enable
<errcode>	Integer type. Error code. See Chapter 6 for more information.

2.3.1.7. AT+QGPSCFG="autogps" Enable/Disable Automatic Running of GNSS

This command enables/disables automatic running of GNSS after the module is powered on, or queries the current settings.

AT+QGPSCFG="autogps" Enable/Disable Automatic Running of GNSS

Write Command AT+QGPSCFG="autogps"[,<autoGPS	Response If the optional parameter is omitted, query the current setting:
---	--

>]	+QGPSCFG: "autogps",<autoGPS> OK If the optional parameter is specified, enable/disable the automatic running of GNSS. OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots. The configuration is saved automatically.

Parameter

<autoGPS>	Integer type. Enable/disable automatic running of GNSS after the module is powered on. 0 Disable automatic running of GNSS. 1 Enable automatic running of GNSS.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

- GNSS runs automatically with a high positioning accuracy and 1 Hz fix rate.
- After enabling this function, BG77xA-GL and BG95xA-GL modules, except BG951A-GL, may fail to register to a network, since they cannot search the network when GNSS is active and in high priority mode.

2.3.1.8. AT+QGPSCFG="priority" Set GNSS or WWAN Priority Mode

This command sets the GNSS or WWAN priority mode or queries the current settings.

AT+QGPSCFG="priority" Set GNSS or WWAN Priority Mode	
Write Command AT+QGPSCFG="priority"[,<priority_type>[,<save>]]	Response If the optional parameters are omitted, query the current setting: +QGPSCFG: "priority",<priority_type>,<state> OK

	If any of the optional parameters is specified, set the GNSS or priority mode: OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. Whether to save the configuration is decided by <save>.

Parameter

<priority_type>	Integer type. Switch between GNSS and WWAN priority mode. 0 GNSS priority mode 1 WWAN priority mode
<save>	Integer type. Choose whether to save the configuration to NVRAM. 0 Do not save to NVRAM 1 Save to NVRAM
<state>	Integer type. GNSS/WWAN state. 0 WWAN/GNSS in unloaded state 1 WWAN in pending state 2 GNSS in pending state 3 WWAN in loaded state 4 GNSS in loaded state
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

BG951A-GL module does not support this command.

2.3.1.9. AT+QGPSCFG="xtrafilesize" Select XTRA File to be Downloaded

This command selects the XTRA file to be downloaded.

AT+QGPSCFG="xtrafilesize" Select XTRA File to be Downloaded

Write Command	Response
AT+QGPSCFG="xtrafilesize" [<type>]	If the optional parameter is omitted, query the current setting: +QGPSCFG: "xtrafilesize",<type> OK

	If the optional parameter is specified, select the XTRA file to be downloaded: OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots. The configuration is saved automatically.

Parameter

<type>	Integer type. XTRA file type. Options for BG77xA-GL and BG95xA-GL, except BG951A-GL: 1 Download 1-day validity CEP file. 2 Download 2-day validity CEP file. 3 Download 3-day validity CEP file. 7 Download 7-day validity CEP file. Options for BG951A-GL: 3 Download 3-day validity LLE file with. 7 Download 7-day validity LLE file. 14 Download 14-day validity LLE file.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

BG951A-GL supports this command only over main AT port when **AT+QGPSCFG="gnss_mode",1**.

2.3.1.10. AT+QGPSCFG="xtra_info" Query Existing XTRA File Information

This command queries existing XTRA file information. You can use the command to manually check XTRA file validity.

AT+QGPSCFG="xtra_info" Query Existing XTRA File Information	
Write Command	Response
AT+QGPSCFG="xtra_info"	+QGPSCFG: "xtra_info",<left_time>,<date>
	OK

	If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	-

Parameter

<left_time>	Integer type. Remaining validity of XTRA file. Range: 0–168. Unit: hour.
<date>	Integer type. Validity start date of XTRA file in UTC format. Format: "YYYY/MM/DD,hh:mm:ss".
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

BG951A-GL supports this command only over main AT port when **AT+QGPSCFG="gnss_mode",1**.

2.3.1.11. AT+QGPSCFG="gpsdop" Query GNSS DOP Value

This command queries GNSS DOP value.

AT+QGPSCFG="gpsdop" Query GNSS DOP Value	
Write Command AT+QGPSCFG="gpsdop"	Response +QGPSCFG: "gpsdop",<PDOP>,<HDOP>,<VDOP> OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	-

Parameter

<PDOP>	Float type. Position dilution of precision.
<HDOP>	Float type. Horizontal dilution of precision.
<VDOP>	Float type. Vertical dilution of precision.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

2.3.1.12. AT+QGPSCFG="nmea_epe" Enable/Disable EPE NMEA Sentence Outputting

This command enables/disables the output of EPE NMEA sentences, which includes the GNSS receiver estimated accuracy.

AT+QGPSCFG="nmea_epe" Enable/Disable EPE NMEA Sentence Outputting	
Write Command AT+QGPSCFG="nmea_epe"[,<NMEA_EPE>]	Response If the optional parameter is omitted, query the current setting. +QGPSCFG: "nmea_epe",<NMEA_EPE> OK If the optional parameter is specified, enable/disable outputting of NMEA message on the receiver's estimated accuracy. OK If there is any error related to ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<NMEA_EPE>	Integer type. Enable/disable EPE NMEA sentence output. 0 Disable 1 Enable Sentence format: \$PQEPE,<timestamp>,<status>,<HDOP>,<hori_unc>,<vert_unc>,<speed_unc>,<head_unc>*<checksum>
<timestamp>	String type. UTC time. Format: "hhmmss.ss"
<status>	String type. Indication of data validity. A Valid V Invalid
<HDOP>	Float type. Horizontal dilution of precision.
<hori_unc>	Float type. Horizontal estimated position error. Unit: meter.
<vert_unc>	Float type. Vertical estimated position error. Unit: meter.
<speed_unc>	Float type. Horizontal estimated velocity error. Unit: m/s.
<head_unc>	Float type. Estimated heading error. Unit: degree.
<checksum>	Hexadecimal type. Checksum is the XOR of all bytes between "\$" and "*" (not including the delimiters themselves).
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

This command is supported only on BG951A-GL.

2.3.1.13. AT+QGPSCFG="estimation_error" Get Estimation Error

This command retrieves the estimation error.

AT+QGPSCFG="estimation_error" Get Estimation Error	
Write Command AT+QGPSCFG="estimation_error"	Response +QGPSCFG: "estimation_error",<hori_unc>,<vert_unc>,<speed_unc>,<head_unc> OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	-

Parameter

<hori_unc>	Float type. Horizontal estimated position error. Unit: meter.
<vert_unc>	Float type. Vertical estimated position error. Unit: meter.
<speed_unc>	Float type. Horizontal estimated velocity error. Unit: m/s.
<head_unc>	Float type. Estimated heading error. Unit: degree.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

2.3.1.14. AT+QGPSCFG="xtra_download" Trigger XTRA File Downloading

This command triggers XTRA file downloading.

AT+QGPSCFG="xtra_download" Trigger XTRA File Downloading	
Write Command AT+QGPSCFG="xtra_download",<type>	Response OK +QGPSURC: "XTRA_DL",<DL_err> If there is any error related to the ME functionality:

	+CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<type>	Integer type. Operating mode. 1 Trigger XTRA file downloading.
<DL_err>	Integer type. Download process result. 0 XTRA file downloaded successfully. -1 Downloading failed.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

BG951A-GL supports this command only over main AT port when **AT+QGPSCFG="gnss_mode",1**.

2.3.1.15. AT+QGPSCFG="xtra_autodownload" Enable/Disable Automatic Downloading of XTRA File

This command enables or disables the automatic downloading of XTRA file upon GNSS starting.

AT+QGPSCFG="xtra_autodownload" Enable/Disable Automatic Downloading of XTRA File	
Write Command AT+QGPSCFG="xtra_autodownload" [,<mode>]	Response If optional parameter is omitted, query the current setting. +QGPSCFG: "xtra_autodownload",<mode> OK If the optional parameter is specified, enable or disable the automatic downloading of XTRA file. OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module is rebooted. The configuration will be saved automatically.

Parameter

<mode>	Integer type. Enable/disable automatic downloading of XTRA file upon GNSS starting. 0 Disable 1 Enable
<errcode>	Integer type. Error code. See Chapter 6 for more information.

2.3.1.16. AT+QGPSCFG="xtra_cfg" Configure XTRA Proxy Server

This command configures the URL, username and password for XTRA (CEP) Proxy Server Connection (see **Chapter 1.4.4.3**).

AT+QGPSCFG="xtra_cfg" Configure XTRA Proxy Server	
Write Command AT+QGPSCFG="xtra_cfg",<URL>[,<username>,<password>]	Response If optional parameter is omitted, query the current setting. +QGPSCFG: "xtra_cfg",<URL>,<username>,<password> OK If the optional parameter is specified, configure the URL, username and password of Proxy Server Connection for downloading XTRA file. OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots.

Parameter

<URL>	String type. URL of target XTRA Proxy Server.
<username>	String type. Username of target XTRA Proxy Server.
<password>	String type. Password of target XTRA Proxy Server.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

2.3.1.17. AT+QGPSCFG="test_mode" Enable/Disable XTRA Test Mode

This command enables/disables XTRA test mode.

AT+QGPSCFG="test_mode" Enable/Disable XTRA Test Mode	
Write Command AT+QGPSCFG="test_mode"[,<mode>]	Response If the optional parameter is omitted, query the current setting: +QGPSCFG: "test_mode",<mode> OK If the optional parameter is specified, enable/disable XTRA test mode. OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots. The configuration is saved automatically.

Parameter

<mode>	Integer type. Enable/disable XTRA test mode. 0 Disable 1 Enable.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

1. When XTRA test mode is enabled, the number of XTRA file downloading attempts is unlimited.
2. XTRA test mode is typically used for testing GNSS cold start performance with the XTRA feature enabled.

2.3.1.18. AT+QGPSCFG="gnss_mode" Select GNSS Mode

This command selects the GNSS mode.

AT+QGPSCFG="gnss_mode" Select GNSS mode	
Write Command AT+QGPSCFG="gnss_mode"[,<mode>]	Response If the optional parameter is omitted, query the current setting.

>]	+QGPSCFG: "gnss_mode",<mode> OK If the optional parameter is specified, select one of GNSS modes. OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots. The configuration is saved automatically.

Parameter

<mode>	Integer type. GNSS mode. <u>0</u> Disable GNSS. 1 GNSS module is used and connected over I2C to modem. GNSS AT commands should be sent over modem's main AT port. 2 GNSS module is used independently of modem. GNSS AT commands should be sent over GNSS UART port.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

This command is available only for BG951A-GL module.

2.3.2. AT+QGPS Turn on GNSS

This command turns on the GNSS function. When **<fix_count>** is 0, the GNSS continuously gets a position fix and is always on. In this case, you can turn off the GNSS with **AT+QGPSEND**. If **<fix_count>** is not 0, the GNSS is turned off automatically when **<fix_count>** reaches the specified value.

AT+QGPS Turn on GNSS

Test Command AT+QGPS=?	Response +QGPS: (list of supported <GNSS_mode>s)[,(list of supported <accuracy>s)[,(list of supported <fix_count>s)[,(list of supported <fix_rate>s)[,(list of supported <HEPE>s)]]]]
---	--

	OK
Read Command AT+QGPS?	Response +QGPS: <GNSS_state>
	OK
Write Command AT+QGPS=<GNSS_mode>[,<accuracy>[,<fix_count>[,<fix_rate>[,<HEPE>]]]]	Response OK [+QGPSURC: "XTRA_DL",<DL_err>] If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<GNSS_state>	Integer type. GNSS state. 0 GNSS OFF 1 GNSS ON 3 GNSS START PENDING 5 GNSS END PENDING
<GNSS_mode>	Integer type. GNSS operating mode. 1 Stand-alone mode
<accuracy>	Integer type. Desired accuracy level acceptable for fix computation. 1 Low Accuracy (1000 m) 2 Medium Accuracy (500 m) 3 High Accuracy (50 m)
<fix_count>	Integer type. Number of positioning or continuous positioning attempts. Range: 0–1000. Default value: 0. 0 indicates continuous positioning. Other values indicate the number of positioning attempts. When the value reaches the specified number of attempts, GNSS will be stopped.
<fix_rate>	Integer type. Interval between the first- and second-time positioning. Unit: second. Range: 1–65535. Default value: 1.
<HEPE>	Integer type. Accuracy threshold. GNSS NMEA sentences will not be output if the accuracy is less than <HEPE> . 0 means disable this function. Range: 0–1000. Default value: 0. Unit: meter.
<DL_err>	Valid on BG951A-GL only. Integer type. XTRA file downloading result. 0 XTRA file downloaded successfully.

	-1 Download failed.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

1. XTRA file downloading is triggered only when XTRA feature is enabled and the XTRA file has expired.
2. **<HEPE>** is supported only on BG951A-GL module. In addition, only the high accuracy level for fix (**<accuracy>=3**) is supported on this module.

2.3.3. AT+QGPSEND Turn off GNSS

This command turns off GNSS. It is used when GNSS is turned on and continuously fixes position (**<fix_count>** is 0). In such a case, GNSS can be turned off with **AT+QGPSEND**.

You do not need to use this command when **<fix_count>** is not 0, as GNSS is turned off automatically when **<fix_count>** reaches the specified value.

AT+QGPSEND Turn off GNSS

Test Command AT+QGPSEND=?	Response OK
Execution Command AT+QGPSEND	Response OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	-

Parameter

<errcode>	Integer type. Error code. See Chapter 6 for more information.
------------------------	--

2.3.4. AT+QGPSLOC Acquire Positioning Information

This command acquires positioning information. Before executing the command, GNSS must be turned on via **AT+QGPS**.

AT+QGPSLOC Acquire Positioning Information	
Test Command AT+QGPSLOC=?	Response +QGPSLOC: (list of supported <mode> s)[,(list of supported <para> s)] OK
Write Command AT+QGPSLOC=<mode>[,<para>]	Response +QGPSLOC: <UTC> , <latitude> , <longitude> , <HDOP> , <altitude> , <fix> , <COG> , <spkm> , <spkn> , <date> , <nsat> OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Read Command AT+QGPSLOC?	Response Return the positioning information with <latitude> , <longitude> in format: ddmm.mmmmN/S,dddmm.mmmmE/W: +QGPSLOC: <UTC> , <latitude> , <longitude> , <HDOP> , <altitude> , <fix> , <COG> , <spkm> , <spkn> , <date> , <nsat> OK
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<mode>	Integer type. Latitude and longitude display format. If <mode> ≤ 2, the timer periodically queries and reports latitude and longitude. The command must be executed only after getting the position fix. 0 <latitude> , <longitude> format: ddmm.mmmmN/S,dddmm.mmmmE/W 1 <latitude> , <longitude> format: ddmm.mmmmmm,N/S,dddmm.mmmmmm,E/W 2 <latitude> , <longitude> format: (-)dd.ddddd,(-)ddd.ddddd If <mode> ≥ 3, the positioning information is reported according to the frequency of outputting RMC NMEA sentences. There is no restriction regarding command execution — the command can be executed at any time. 3 <latitude> , <longitude> format: ddmm.mmmmN/S,dddmm.mmmmE/W
---------------------	---

<para>	4	<latitude>,<longitude> format: ddmm.mmmmmm,N/S,dddmm.mmmmmm,E/W
	5	<latitude>,<longitude> format: (-)dd.ddddd,(-)ddd.ddddd
<para>		Integer type.
		If <mode> <=2, the parameter specifies the time when the queried results are reported. Range: 0–3600. Default value: 0. Unit: second.
<para>	0	Disable reporting.
		If <mode> >=3, the parameter specifies positioning information reporting mode. Range: 0–2.
<para>	0	Disable reporting.
	1	Report only once when fix status changes: from "not fixed" to "fixed", or vice versa.
<para>	2	Always report. Report according to NMEA sentence output frequency.
		String type. Current UTC time.
<UTC>		Format: hhmmss.sss (Quoted from GPGLL sentence).
<latitude>		Float type. Latitude.
		If <mode> is 0 or 3:
<latitude>		Format: ddmm.mmmmmN/S (Quoted from GPGLL sentence)
	dd	00–89 (Unit: degree)
<latitude>	mm.mmmmm	00.0000–59.9999 (Unit: minute)
	N/S	North latitude / South latitude
<latitude>		If <mode> is 1 or 4:
		Format: ddmm.mmmmmm,N/S (Quoted from GPGLL sentence)
<latitude>	dd	00–89 (Unit: degree)
	mm.mmmmmm	00.000000–59.999999 (Unit: minute)
<latitude>	N/S	North latitude / South latitude
		If <mode> is 2 or 5:
<latitude>		Format: (-)dd.ddddd (Quoted from GPGLL sentence)
	dd.ddddd	-89.99999 to 89.99999 (Unit: degree)
<latitude>	-	South latitude
		Float type. Longitude.
<longitude>		If <mode> is 0 or 3:
		Format: dddmm.mmmmmE/W (Quoted from GPGLL sentence)
<longitude>	ddd	000–179 (Unit: degree)
	mm.mmmmm	00.0000–59.9999 (minute)
<longitude>	E/W	East longitude / West longitude
		If <mode> is 1 or 4:
<longitude>		Format: dddmm.mmmmmm,E/W (Quoted from GPGLL sentence)
	ddd	000–179 (Unit: degree)
<longitude>	mm.mmmmmm	00.000000–59.999999 (Unit: minute)
	E/W	East longitude / West longitude
<longitude>		If <mode> is 2 or 5:
		Format: (-)dd.ddddd (Quoted from GPGLL sentence)
	dd.ddddd	-179.99999 to 179.99999 (Unit: degree)
<longitude>	-	West longitude

<HDOP>	Float type. Horizontal precision. Range: 0.5–99.9 (Quoted from GPGLL sentence).
<altitude>	Float type. Antenna altitude above mean sea level, accurate to one decimal place. Unit: meter. (Quoted from GPGLL sentence)
<fix>	Integer type. GNSS positioning mode (Quoted from GNGGA/GPGGA sentence). 2 2D positioning 3 3D positioning
<COG>	String type. Course over ground based on true north. Format: ddd.mm (Quoted from GPVTG sentence). ddd 000–359 (Unit: degree) mm 00–59 (Unit: minute)
<spkm>	Float type. Speed over ground. Format: xxxx.x. Unit: km/h. Accurate to one decimal place (Quoted from GPVTG sentence).
<spkn>	Float type. Speed over ground. Format: xxxx.x. Unit: knots. Accurate to one decimal place (Quoted from GPVTG sentence).
<date>	String type. UTC time when fixing a position. Format: ddmmyy (Quoted from GPRMC sentence).
<nsat>	Integer type. Number of satellites. Range: 00–12 (Quoted from GPGLL sentence).
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

The response of **AT+QGPSLOC?** is the same as that of **AT+QGPSLOC=0**.

2.3.5. AT+QGPSGNMEA Acquire NMEA Sentences

This command acquires NMEA sentences. Before executing this command, turn on GNSS via **AT+QGPS**, and set <NMEA_src> of **AT+QGPSCFG="nmeasrc"** to 1.

Sentence output can also be disabled via **AT+QGPSCFG="gpsnmeatype",0**, **AT+QGPSCFG="glossnmeatype",0**.

AT+QGPSGNMEA Acquire NMEA Sentences

Test Command AT+QGPSGNMEA=?	Response +QGPSGNMEA: (list of supported <NMEA_sentence>s) OK
Write Command Acquire GGA sentences AT+QGPSGNMEA="GGA"	Response +QGPSGNMEA: <GGA_sentence> OK

	If there is any error related to the ME functionality: +CME ERROR: <errcode>
Write Command Acquire RMC sentences AT+QGPSGNMEA="RMC"	Response +QGPSGNMEA: <RMC_sentence> OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Write Command Acquire GSV sentences AT+QGPSGNMEA="GSV"	Response +QGPSGNMEA: <GSV_sentence> OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Write Command Acquire GSA sentences AT+QGPSGNMEA="GSA"	Response +QGPSGNMEA: <GSA_sentence> OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Write Command Acquire VTG sentences AT+QGPSGNMEA="VTG"	Response +QGPSGNMEA: <VTG_sentence> OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	-

Parameter

<GGA_sentence>	GGA sentences.
<RMC_sentence>	RMC sentences.
<GSV_sentence>	GSV sentences.
<GSA_sentence>	GSA sentences.
<VTG_sentence>	VTG sentences.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

2.3.6. AT+QCFGEXT Query and Configure Extended Settings

This command queries and configures various extended module settings.

AT+QCFGEXT Query and Configure Extended Settings	
Test Command AT+QCFGEXT=?	Response +QCFGEXT: "addgeo",<geoid>,<mode>,<shape>,<lat1>,<lon1>,<lat2>,<lon2>,<lat3>,<lon3>,<lat4>,<lon4>]]] +QCFGEXT: "deletegeo",<geoid> +QCFGEXT: "querygeo",<geoid> OK
Maximum Response Time	300 ms
Characteristics	-

2.3.6.1. AT+QCFGEXT="addgeo" Add a Geo-fence

This command adds a geo-fence or queries the current settings.

AT+QCFGEXT="addgeo" Add a Geo-fence	
Write Command AT+QCFGEXT="addgeo",<geoid>,<mode>,<shape>,<lat1>,<lon1>,<lat2>,<lon2>,<lat3>,<lon3>,<lat4>,<lon4>]]]]	Response If all optional parameters are omitted, query the current settings of all added geo-fences. [+QCFGEXT: "addgeo",<geoid>,<mode>,<shape>,<lat1>,<lon1>,<lat2>,<lon2>,<lat3>,<lon3>,<lat4>,<lon4>]]] ... +QCFGEXT: "addgeo",<geoid>,<mode>,<shape>,<lat1>,<lon1>,<lat2>,<lon2>,<lat3>,<lon3>,<lat4>,<lon4>]]] OK If the optional parameters after <geoid> are omitted, query the current setting of the specified geo-fence. +QCFGEXT: "addgeo",<geoid>,<mode>,<shape>,<lat1>,<lon1>,<lat2>,<lon2>,<lat3>,<lon3>,<lat4>,<lon4>]]] OK If <shape>=0, add a circular geo-fence and the parameters after <lat2> must be omitted. OK

	If <shape>=1, add a circular geo-fence and the parameters after <lon2> must be omitted. OK If <shape>=2, add a triangular geo-fence and the parameters after <lon3> must be omitted. OK If <shape>=3, add a quadrangle geo-fence. All parameters must be specified. OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<geoid>	Integer type. Geo-fence ID. Range: 0–9.
<mode>	Integer type. URC reporting mode. 0 Disable URC reporting when entering or leaving the geo-fence 1 Enable URC reporting when entering the geo-fence 2 Enable URC reporting when leaving the geo-fence 3 Enable URC reporting when entering or leaving the geo-fence The URC is shown below: +QIND: "GEOFENCE",<ID>,<action>,<time>,<latitude>,<longitude>,<altitude>,<course>,<speed>,<PDOP>,<HDOP>,<VDOP> URC parameters: <ID> ID of geo-fence to be entered or exited. <action> Current action of the module. 1 Entering the geo-fence 2 Leaving the geo-fence <time> UTC time when entering or exiting the geo-fence. Format: yyyy-MM-dd hh:mm:ss <latitude> Latitude of module when entering or leaving the geo-fence. Unit: degree. Format: ±dd.dddddd. Range: -90.000000 to 90.000000. <longitude> Longitude of module when entering or leaving the geo-fence. Unit: degree. Format: ±ddd.dddddd. Range: -180.000000 ° to 180.000000 °.

	<altitude>	Antenna altitude above mean sea level. Unit: meter.
	<course>	Course over ground, relative to true north. Unit: degree.
	<speed>	Speed over ground. Unit: m/s.
	<PDOP>	Position dilution of precision.
	<HDOP>	Horizontal dilution of precision.
	<VDOP>	Vertical dilution of precision.
	<shape>	Integer type. Geo-fence shape. 0 Circularity with center and radius 1 Circularity with center and one point on the circle 2 Triangle 3 Quadrangle
	<lat1>	Latitude of a point defined as the center of the geo-fence circular area or the first point. Unit: degree. Format: $\pm dd.ddddddd$. Range: -90.000000° to 90.000000° .
	<lon1>	Longitude of a point defined as the center of the geo-fence circular area or the first point. Unit: degree. Format: $\pm ddd.ddddddd$. Range: -180.000000 to 180.000000 .
	<lat2>	When <shape> is 0, this parameter is a radius. Range: $0-6000000$. Unit: meter. When <shape> is not 0, this parameter is a latitude. Unit: degree. Format: $\pm dd.ddddddd$. Range: -90.000000 to 90.000000 . If <shape> is 0, the parameters after <lat2> must be omitted.
	<lon2>	Longitude of the second point. Unit: degree. Format: $\pm ddd.ddddddd$. Range: -180.000000 to 180.000000 . If <shape> is 1, the parameters after <lon2> must be omitted.
	<lat3>	Latitude of the third point. Unit: degree. Format: $\pm dd.ddddddd$. Range: -90.000000 to 90.000000 .
	<lon3>	Longitude of the third point. Unit: degree. Format: $\pm ddd.ddddddd$. Range: -180.000000 to 180.000000 . If <shape> is 2, the parameters after <lon3> must be omitted.
	<lat4>	Latitude of the fourth point. Unit: degree. Format: $\pm dd.ddddddd$. Range: -90.000000 to 90.000000 .
	<lon4>	Longitude of the fourth point. Unit: degree. Format: $\pm ddd.ddddddd$. Range: -180.000000 to 180.000000 .
	<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

BG951A-GL supports only up to 5 **<geoid>**s. Geo-fence **<shape>** can only be defined as circularity with center and radius.

2.3.6.2. AT+QCFGEXT="deletegeo" Delete a Geo-fence

This command deletes a geo-fence.

AT+QCFGEXT="deletegeo" Delete a Geo-fence	
Write Command AT+QCFGEXT="deletegeo",<geoid>	Response: OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<geoid>	Integer type. Geo-fence ID. Range: 0–10. If 10, dele all geo-fences.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

BG951A-GL supports only up to 5 **<geoid>**s. ID 5 means deleting all geo-fences.

2.3.6.3. AT+QCFGEXT="querygeo" Query Position Relative to Geo-Fence

This command queries the position relative to the geo-fence.

AT+QCFGEXT="querygeo" Query Position Relative to Geo-Fence	
Write Command AT+QCFGEXT="querygeo",<geoid>	Response: +QCFGEXT: "querygeo",<geoid>,<pos_wrt_geofence> OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	-

Parameter

<geoid>	Integer type. Geo-fence ID. Range: 0–9.
<pos_wrt_geofence>	Integer type. Position relative to geo-fence. 0 Position unknown 1 Position inside the geo-fence 2 Position outside the geo-fence
<errcode>	Integer type. Error code. See Chapter 6 for more information.

2.3.7. AT+QGPSXTRA Enable/Disable XTRA Function

This command enables/disables the XTRA function.

AT+QGPSXTRA Enable/Disable XTRA Function	
Test Command AT+QGPSXTRA=?	Response +QGPSXTRA: (list of supported <XTRA_enable>s) OK
Read Command AT+QGPSXTRA?	Response +QGPSXTRA: <XTRA_enable> OK
Write Command AT+QGPSXTRA=<XTRA_enable>	Response OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots. The configuration is saved automatically.

Parameter

<XTRA_enable>	Integer type. Enable/disable the XTRA feature. 0 Disable 1 Enable
<errcode>	Integer type. Error code. See Chapter 6 for more information.

2.3.8. AT+QGPSXTRATIME Inject XTRA Time

This command injects XTRA time to the GNSS engine when the module cannot synchronize time from network. Before using this command, enable XTRA feature via **AT+QGPSXTRA=1**.

AT+QGPSXTRATIME Inject XTRA Time	
Test Command AT+QGPSXTRATIME=?	Response +QGPSXTRATIME: 0,<XTRA_time>,(list of supported <UTC>s) OK
Read Command AT+QGPSXTRATIME?	Response +QGPSXTRATIME: <inject_time> OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Write Command AT+QGPSXTRATIME=<op>,<XTRA_time>[,<UTC>]	Response OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	-

Parameter

<inject_time>	String type. Injected UTC time. Format: "YYYY/MM/DD,hh:mm:ss". e.g., "2020/04/03,15:34:50". If the queried <inject_time> is empty, it means that XTRA time has not been injected yet.
<op>	Integer type. Operation type. 0 Inject XTRA time
<XTRA_time>	String type. Current UTC time to be injected. Format: "YYYY/MM/DD,hh:mm:ss". e.g., "2020/04/03,15:34:50".
<UTC>	Integer type. Type of time. 0 Reserved <u>1</u> UTC time
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

The module automatically synchronizes time from network by NITZ and supports querying the injected time via **AT+QGPSXTRATIME?**. If the XTRA time is not injected, you can inject it manually with this command.

2.3.9. AT+QGPSXTRADATA Inject Assistance Data to GNSS Engine

This command injects assistance data to the GNSS engine.

AT+QGPSXTRADATA Inject Assistance Data to GNSS Engine	
Test Command AT+QGPSXTRADATA=?	Response +QGPSXTRADATA: <XTRA_data_filename> OK
Read Command AT+QGPSXTRADATA?	Response +QGPSXTRADATA: <XTRA_data_durtime>,<injected_datatime> OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Write Command AT+QGPSXTRADATA=<XTRA_data_filename>	Response OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<XTRA_data_filename>	String type. Filename of XTRA file, e.g., "lle_gps.lle"
<XTRA_data_durtime>	Integer type. Validity period of injected XTRA file. Unit: hour. 24 1-day XTRA file 48 2-day XTRA file 72 3-day XTRA file 168 7-day XTRA file 336 14-day XTRA file
<injected_datatime>	String type. Start of XTRA file validity period.

<errcode>	Format: "YYYY/MM/DD,hh:mm:ss", e.g., "2021/07/20,06:00:00". Integer type. Error code. See Chapter 6 for more information.
------------------------	---

NOTE

The XTRA file expires at the end of validity period: **<injected_datatime> + <XTRA_data_durtime>**, but this period can also be slightly shorter, depending on the time when XTRA file has been updated on the server side. To check the remaining time in hours, execute **AT+QGPSCFG="xtra_info"**.

2.3.10. AT+QGPSDEL Delete Assistance Data

This command deletes assistance data so as to perform a cold start, hot start and warm start of GNSS. The command can only be executed when GNSS is turned off. After deleting the assistance data via this command, GNSS cold start can be initiated via **AT+QGPS**. Hot/warm start can also be performed if the relevant requirements are met.

AT+QGPSDEL Delete Assistance Data	
Test Command AT+QGPSDEL=?	Response +QGPSDEL: (list of supported <delete_type>) OK
Write Command AT+QGPSDEL=<delete_type>	Response OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
The maximum response time	1 s
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<delete_type>	Integer type. Type of GNSS assistance data to be deleted. 0 Delete all assistance data including XTRA data. Enforce cold start after starting GNSS. 1 Do not delete any data. After starting GNSS, perform hot start if conditions have been met. 2 Delete some related data. After starting GNSS, perform warm start if conditions have been met.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

When performing a cold start, the XTRA file is deleted and cannot be downloaded again for the next 24 hours. You can execute **AT+QGPSCFG="test_mode",1** to remove the download limitation. For more information, see **Chapter 2.3.1.17**. Currently, there is no download limitation.

2.4. BG951A-GL-Specific AT Command Description

GNSS AT commands defined in this chapter are specific only for BG951A-GL GNSS engine. BG770A-GL and BG950A-GL do not support these commands.

2.4.1. AT+QGPSADATA Query and Configure LLE Data

This command queries and configures LLE data (GNSS assistance data). LLE data files for each GNSS constellation can be acquired from LLE server via the Internet. LLE data are stored in the flash memory connected to the BG951A-GL GNSS engine by injecting them with **AT+QGPSADATA?**. Since **AT+QGPSADATA?** returns injection result, it can be used to check LLE data validity.

AT+QGPSADATA Query and Configure LLE Data	
Test Command AT+QGPSADATA=?	Response +QGPSADATA:"erase" +QGPSADATA:"write",0,(list of supported <serial_number>s),(list of supported <size>s) OK
Read Command AT+QGPSADATA?	Response +QGPSADATA:GPS,<start_time>,<duration> +QGPSADATA:QZSS,<start_time>,<duration> +QGPSADATA:GLONASS,<start_time>,<duration> +QGPSADATA:Galileo,<start_time>,<duration> +QGPSADATA:BeiDou,<start_time>,<duration> OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	-
Characteristics	-

Parameter

<start_time>	String type. Start time (UTC) of each piece of LLE data. Format: "YYYY/MM/DD hh:mm". e.g. "2021/04/03 00:00" If LLE data do not exist, "0000/00/00 00:00" is returned.
<duration>	Integer type. Validity in days as of injection time. If LLE data do not exist, "0" returns.
<errcode>	Integer type. Error code. See Chapter 6 for more information.
<serial_number>	See Chapter 2.4.1.2 for more information.
<size>	See Chapter 2.4.1.2 for more information.

2.4.1.1. AT+QGPSADATA="erase" Erase LLE Data

This command erases LLE data that are already written in the BG951A-GL GNSS engine. LLE data for all constellations are erased by this command.

AT+QGPSADATA="erase" Erase LLE Data	
Write Command AT+QGPSADATA="erase"	Response OK If there is any error: +CME ERROR: <errcode>
Maximum Response Time	-
Characteristics	-

2.4.1.2. AT+QGPSADATA="write" Write LLE Data

This command writes the LLE data to the BG951A-GL GNSS engine. LLE data files must be divided in 2 KB packets (the last packet will be a fraction) and transferred. The prefix "0xA0" must be added at the start of each packet. One packet is sent with one command. The serial number of the packet and the data size are specified in the arguments.

AT+QGPSADATA="write" Write LLE Data	
Write Command AT+QGPSADATA="write",<hash>,<serial_number>,<size>	Response OK Following the 1st "OK", inject the LLE data packet (2 KB data or fraction + 1 byte prefix) from the host controller. OK

	If there is any error: +CME ERROR: <errcode>
Maximum Response Time	-
Characteristics	-

Parameter

<hash>	N/A. Set "0" always.
<serial_number>	Integer type. Packet serial number. Range: 0-127.
<size>	Integer type. Packet chunk size. Unit: Byte. Range: 1-2048.
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

There is no need to erase the LLE data that have already been written in the BG951A-GL GNSS engine since LLE data for each GNSS constellation are overwritten.

2.4.2. AT+QGPSRM Enable/Disable RF Mute

This command enables/disables RF mute functionality. RF mute functionality is used for reducing the GNSS time to first fix, so that GNSS engine would get a position fix faster. If RF part is disabled (RF mute is enabled), ephemeris data will be checked only once, which will reduce the time to first fix.

AT+QGPSRM Enable/Disable RF Mute	
Test Command AT+QGPSRM=?	Response +QGPSRM:(list of supported <RF_mute>s) OK
Write Command AT+QGPSRM=<RF_mute>	Response OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	
Characteristics	

Parameter

<RF_mute>	Integer type. Enable/disable RF mute functionality. 0 Disable
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	<u>1</u> Enable
<errcode>	Integer type. Error code. See Chapter 6 for more information.

2.4.3. AT+QGPSPPS Enable/Disable 1PPS Output

This command enables/disables 1PPS output.

AT+QGPSPPS Enable/Disable 1PPS Output	
Test Command AT+QGPSPPS=?	Response +QGPSPPS:(list of supported <ppsout>s) OK
Write Command AT+QGPSPPS=<ppsout>	Response OK If there is any error related to the ME functionality: +CME ERROR: <errcode>
Maximum Response Time	-
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<ppsout>	Integer type. Enable/disable 1PPS output. <u>0</u> Disable 1 Enable
<errcode>	Integer type. Error code. See Chapter 6 for more information.

NOTE

When 1PPS output is disabled, SFNIND_1PPS pin will be set to Hi-Z.

2.4.4. AT+QGSRST Reset GNSS

This command resets the BG951A-GL GNSS engine. 1 s after this command is issued, the engine is rebooted and ready to receive the commands.

AT+QGSRST Reset GNSS	
Test Command AT+QGSRST=?	Response OK

Execution Command AT+QGPSRST	Response OK If there is any error: +CME ERROR: <errcode>
Maximum Response Time	-
Characteristics	-

Parameter

<errcode> Integer type. Error code. See **Chapter 6** for more information.

NOTE

BG951A-GL GNSS engine reboots in 1.2 s. Issue next command after at least 1.2 s of receiving **OK** response.

2.4.5. AT+QGPSPSM Sleep and Wake up GNSS

This command puts the BG951A-GL GNSS engine to sleep or wakes it up.

AT+QGPSPSM Sleep and Wake up GNSS	
Test Command AT+QGPSPSM=?	Response +QGPSPSM:(list of supported <operation>s) OK
Write Command AT+QGPSPSM=<operation>	Response OK If there is any error: +CME ERROR: <errcode>
Maximum Response Time	-
Characteristics	-

Parameter

String type. Write command operating mode.		
<operation>	"SLEEP0"	Put BG951A-GL GNSS engine to sleep mode 0
	"SLEEP1"	Put BG951A-GL GNSS engine to sleep mode 1

"WAKEUP" Wake up BG951A-GL GNSS engine

<errcode> Integer type. Error code. See **Chapter 6** for more information.

Example

```
AT+QGPS=1           //Turn on GNSS.
OK

AT+QGPSEND          //Turn off GNSS.
OK

AT+QGPSPSM="SLEEP0" //GNSS engine goes to sleep.
OK

AT+QGPSPSM="WAKEUP" //GNSS engine wakes up.
ERROR

AT+QGPSPSM="WAKEUP" //Any command can be sent because GNSS engine is awake now.
OK
```

NOTE

1. To ensure successful execution of **AT+QGPSPSM**, BG951A-GL GNSS engine must be turned off before you execute the command.
2. After BG951A-GL GNSS engine enters sleep mode, if you run **AT+QGPSPSM="WAKEUP"** it will fail. The **ERROR** response will activate the connection between the GNSS engine and modem, after which running **AT+QGPSPSM="WAKEUP"** again will succeed and any other command can be executed properly.

3 Examples

3.1. Turn on/off GNSS

Default arguments are used in this example to turn on GNSS. Once GNSS is turned on, NMEA sentences will be outputted from "usbmea" port by default, and GNSS can be turned off via **AT+QGPSEND**.

```
AT+QGPS=1           //Turn on GNSS.
OK

//After turning on GNSS, NMEA sentences are outputted from "usbmea" port by default.
AT+QGPSLOC?         //Obtain positioning information.
+QGPSLOC: 130618.000,3150.8076N,11711.9039E,0.8,89.5,2,0.00,0.0,0.0,110919,12

OK
AT+QGPSEND          //Turn off GNSS.
OK
```

3.2. Acquire Positioning Information

The following example illustrates how to acquire positioning information via **AT+QGPSLOC** after turning on GNSS and obtaining the position fix.

```
AT+QGPS=1           //Turn on GNSS.
OK
AT+QGPSLOC?
+CME ERROR: 516      //Not fixed
AT+QGPSLOC?         //Obtain positioning information.
+QGPSLOC: 130618.000,3150.8076N,11711.9039E,0.8,89.5,2,0.00,0.0,0.0,110919,12

OK
AT+QGPSLOC=0        //Set the latitude and longitude display format to
                    //ddmm.mmmmN/S,dddmm.mmmmE/W
+QGPSLOC: 131050.000,3150.8069N,11711.9032E,1.2,90.7,3,0.00,0.0,0.0,110919,08
```


OK

AT+QGPSLOC=1 //Set the latitude and longitude display format to
ddmm.mmmmmm,N/S,dddmm.mmmmmm,E/W

+QGPSLOC: 131117.000,3150.806972,N,11711.903278,E,1.3,90.6,3,0.00,0.0,0.0,110919,07

OK

AT+QGPSLOC=2 //Set the latitude and longitude display format to
(-)dd.ddddd,(-)ddd.ddddd

+QGPSLOC: 131140.000,31.84678,117.19838,1.3,90.5,3,0.00,0.0,0.0,110919,07

OK

AT+QGPSLOC=2,1 //Obtain positioning information and enable periodical location reporting.

+QGPSLOC: 131305.000,31.84678,117.19838,1.8,89.9,3,0.00,0.0,0.0,110919,07

OK

+QGPSLOC: 131306.000,31.84678,117.19838,1.0,89.9,3,0.00,0.0,0.0,110919,08

+QGPSLOC: 131307.000,31.84678,117.19838,1.0,89.9,3,0.00,0.0,0.0,110919,08

+QGPSLOC: 131308.000,31.84678,117.19838,0.9,89.9,3,0.00,0.0,0.0,110919,08

AT+QGPSLOC=2,0 //Obtain positioning information and disable periodical location reporting.

+QGPSLOC: 131431.000,31.84678,117.19838,0.9,89.7,3,0.00,0.0,0.0,110919,09

OK

AT+QGPSLOC=4,1 //Set position reporting when the positioning status changes.

OK

//Not fixing → fixing

+QGPSLOC:052943.000,3149.314079,N,11706.979269,E,0.8,128.7,3,0.00,0.0,0.0,080720,07

//Fixing → not fixing

+QGPSLOC: ,,,,,,,,,,

AT+QGPSLOC=4,2 //Set location information reporting according to the reporting
frequency of NMEA sentences

+QGPSLOC: 053018.000,3149.309670,N,11706.977340,E,0.8,136.6,3,1.47,2.2,1.2,080720,07

+QGPSLOC: 053019.000,3149.309125,N,11706.978666,E,0.7,136.0,3,0.00,0.0,0.0,080720,08

+QGPSLOC: 053020.000,3149.309125,N,11706.978666,E,0.7,136.0,3,0.00,0.0,0.0,080720,08

```
AT+QGPSLOC=4,0 //Disable location reporting.
OK
```

3.3. Query Satellite System

```
AT+QGPSCFG="gnssconfig" //Query enabled satellite systems.
+QGPSCFG: "gnssconfig",1 //GPS and GLONASS are enabled.

OK
AT+QGPS=1 //Turn on GNSS.
OK
AT+QGPSSNMEA="GSV"

//GPS GSV sentences
+QGPSSNMEA: $GPGSV,4,1,14,02,68,055,16,04,00,000,31,05,64,314,26,07,08,070,18,1*6D
+QGPSSNMEA: $GPGSV,4,2,14,09,04,037,30,12,11,228,36,15,16,207,20,19,11,155,14,1*6B
+QGPSSNMEA: $GPGSV,4,3,14,25,09,269,20,29,27,316,32,06,26,101,,13,41,177,,1*68
+QGPSSNMEA: $GPGSV,4,4,14,17,00,000,,30,10,091,,1*6D

//GLONASS GSV sentences
+QGPSSNMEA: $GLGSV,2,1,05,22,32,332,26,20,25,136,13,21,78,073,18,07,57,243,21,1*78
+QGPSSNMEA: $GLGSV,2,2,05,08,10,224,,1*40

OK
```

3.4. Implement Geo-Fence Function

```
AT+QCFGEXT="addgeo",0,3,0,31.826,117.2168,100 //Add a circular geo-fence 0.
OK
AT+QCFGEXT="addgeo",0 //Query the setting of circular geo-fence 0.
+QCFGEXT: "addgeo",0,3,0,31.826000,117.216800,100.0

OK
AT+QCFGEXT="addgeo",7,1,3,31.833348,117.212909,31.826453,117.213248,31.82873,117.222093,
31.833502,117.220862 //Add a quadrangle geo-fence 7.
OK
AT+QCFGEXT="addgeo",7 //Query the settings of geo-fence 7.
+QCFGEXT:
"addgeo",7,1,3,31.833348,117.212909,31.826453,117.213248,31.828730,117.222093,31.833502,117.
220862
```

```

OK
AT+QCFGEXT="deletegeo",7           //Delete geo-fence 7.
OK
AT+QGPS=1                           //Turn on the GNSS engine.
OK
AT+QCFGEXT="querygeo",0             //Query the position with respect to geo-fence 0.
+QCFGEXT: "querygeo",0,1           //The current position is inside the geo-fence 0.

OK

//When entering geo-fence 0, this URC is reported.
+QIND: "GEOFENCE",0,1,17/08/25 08:35:53,31.825179,117.217127,34.0,0.2,13.8,1.1,0.7,0.8
//When leaving geo-fence 0, this URC is reported.
+QIND: "GEOFENCE",0,2,17/08/25 08:36:07,31.826951,117.217071,38.0,359.0,13.4,0.9,0.6,0.6

```

3.5. How to Use XTRA Feature

There are two ways to download XTRA files:

- a) Via AT command
- b) By starting GNSS

3.5.1. XTRA File Download is Triggered with AT+QGPSCFG="xtra_download",1

```

AT+QGPSXTRA=1           //Enable XTRA and then restart the module to activate the feature.
OK
AT+CEREG?               //Check network status.
+CEREG: 0,1

OK
AT+QGPSXTRATIME?        //Check XTRA time. Inject XTRA time if it is not available.
+QGPSXTRATIME: "2020/07/20,02:34:47"

OK
AT+QGPSCFG="xtra_info"  //Check whether the existing XTRA file has expired.
+QGPSCFG: "xtra_info",0, "2020/07/16,06:00:00" //XTRA file has expired.

OK
AT+QGPSCFG="xtra_download",1 //Trigger XTRA file downloading.
OK

```

```
+QGPSURC: "XTRA_DL",0           //XTRA file has been downloaded successfully.
AT+QGPSXTRADATA?                //Check the validity of the existing XTRA file.
+QGPSXTRADATA: 4320,"2020/07/20,06:00:00"

OK
AT+QGPS=1                       //Start GNSS with XTRA.
OK
```

3.5.2. XTRA File Downloading Triggered by Starting GNSS

```
//Enable XTRA file downloading by starting GNSS, then perform the following procedures.
AT+QGPSXTRA=1                   // Enable XTRA and then restart the module to activate the feature.
OK
AT+CEREG?                       //Check network status.
+CEREG: 0,1

OK
AT+QGPSXTRATIME?                //Check XTRA time. Inject XTRA time if it is not available.
+QGPSXTRATIME: "2020/07/20,02:34:47"

OK
AT+QGPSCFG="xtra_info"          //Check whether the existing XTRA file has expired.
+QGPSCFG: "xtra_info",0, "2020/07/16,06:00:00" //XTRA file has expired.

OK
AT+QGPS=1                       //First, download the XTRA file. After successful downloading, start the GNSS.
OK

+QGPSURC: "XTRA_DL",0

AT+QGPSXTRADATA?                //Check the validity of the existing XTRA file.
+QGPSXTRADATA: 4320,"2020/07/20,06:00:00" //The XTRA file expires on 2020/07/23,06:00:00.

OK
```

4 Use Cases

Considering GNSS feature complexity and the fact that the use cases listed below cannot cover all tracking applications and extreme use cases, we strongly suggest that you provide the specific application cases to Quectel Technical Support team for review via support@quectel.com or <http://e-service.quectel.com>.

4.1. Enable PSM

The recommended procedure for using the PSM is given below. For more information on PSM implementation, see **document [3]**.

Procedure

1. Power on the module and then register to a network.
2. Configure PSM (such as T3412 = 6 hours, T3324 = 10 minutes).
3. Wake up from PSM.
4. Set the module into GNSS priority mode.
5. Turn on GNSS and get a position fix.
6. Switch to WWAN priority mode.
7. Report positioning information and other data to the server.
8. Enter PSM.
9. Repeat 3-8.

4.2. Recommended Mode: WWAN Priority Mode

WWAN priority mode is recommended in case of a high demand for data transmission services. Before using WWAN priority mode, make sure that both requirements below are satisfied:

- eDRX is active, and the eDRX cycle is longer than TTFF.
- There is a low demand for GNSS positioning.

The recommended procedure for using WWAN priority mode is given below.

Procedure

1. Configure the eDRX cycle that is sufficiently longer than TTFF ¹⁾
2. Wake up from sleep mode.
3. Turn on GNSS (GNSS session may be deferred to eDRX).
4. Obtain GNSS positioning information.
5. Turn off GNSS or switch to the WWAN priority mode².
6. Report positioning information to the server.
7. Return to sleep mode.
8. Repeat 2-7.

NOTE

1. ¹⁾ For more information on TTFF values, refer to the corresponding hardware design documents.
2. Step 5 is optional, since it depends on the practical applications and the current consumption requirements.

4.3. Recommended Mode: GNSS Priority Mode

GNSS priority mode is recommended in case of a high demand for GNSS positioning and low demand for data transmission.

The recommended procedure for using GNSS priority mode.

Procedure

1. Wake up from sleep mode.
2. Turn on GNSS (GNSS session starts immediately).
3. Collect data from sensors and obtain GNSS positioning information.
4. Turn off GNSS or switch to WWAN priority mode.
5. Report sensor data and positioning information to the server.
6. Return to sleep mode.
7. Repeat 1–6.

4.4. GNSS Limitations

If the reporting is performed following the steps below, there may be limitations on reporting positioning and sensor data. Success of the query at step 7 cannot be guaranteed every time, because the page may be lost when GNSS is in active status with GNSS priority. In addition, getting positioning information on

time cannot be guaranteed either if WWAN is preferential. In these cases, it is recommended to select a module on which GNSS and WWAN can work simultaneously, either Quectel BG951A-GL, or BG770A-GL or BG950A-GL module with an external GNSS module.

Procedure

1. Wake up from sleep mode.
2. Turn on GNSS (GNSS session started immediately).
3. Collect data from sensors and obtain GNSS positioning information.
4. Turn off GNSS or switch to WWAN priority mode.
5. Report sensor data and positioning information to server.
6. Return to sleep mode.
7. The sensor data and positioning information may be queried from the network side anytime during steps 2–5.
8. Repeat steps 1–7.

5 FAQ

1. **Q:** Why can't the module register to a network when the GNSS auto-start feature is enabled with **AT+QGPSCFG="autogps",1**?
A: If your module is in GNSS priority mode, if GNSS starts automatically when the module powers up, the module will fail to register to a network. Please configure the module into WWAN priority mode if GNSS auto-start feature is needed.
2. **Q:** When GNSS is active and in high priority, the module can detach from the network after executing **AT+CFUN=0**, but why can't it register on the network after executing **AT+CFUN=1**?
A: Network detachment has higher priority than GNSS/WWAN, so the module can detach from the network even if the module is in GNSS priority mode. However, network attachment does not have the same priority as network detachment, so the module cannot register to the network if it is in GNSS priority mode.
3. **Q:** Why reporting sensor data and positioning information sometimes does not work?
A: Reporting sensor data and positioning information cannot be guaranteed every time because the page may be lost when GNSS is in the active status with GNSS priority. In addition, getting positioning information on time cannot be guaranteed if WWAN is preferential. In these cases, it is recommended to select a module on which GNSS and WWAN can work simultaneously, either Quectel BG951A-GL, or BG770A-GL or BG950A-GL module with an external GNSS module.

6 Summary of Error Codes

The **<errcode>** indicates an error related to GNSS operation. The details about **<errcode>** are described in the following table.

Table 6: Summary of Error Codes

<errcode>	Description
501	Invalid parameter
502	Operation not supported
503	GNSS subsystem busy
504	Session is ongoing
505	Session not active
506	Operation timeout
507	Function not enabled
508	Time information error
509	XTRA not enabled
512	Validity time is out of range
513	Internal resource error
514	GNSS locked
515	End by E911
516	No fix
517	Geo-fence ID does not exist
518	Sync time failed
519	XTRA file does not exist

520	XTRA file on downloading
521	XTRA file is valid
522	GNSS is working
523	Time injection error
524	XTRA file is invalid
549	Unknown error

7 Appendix References

Table 7: Related Documents

Document Name
[1] Quectel_BG77xA-GL&BG95xA-GL&BG950S-GL_AT_Commands_Manual
[2] Quectel_BG77xA-GL&BG95xA-GL_GNSS_Jamming_Detection_Application_Note
[3] Quectel_BG77xA-GL&BG95xA-GL&BG950S-GL_PSM_Application_Note

Table 8: Terms and Abbreviations

Abbreviation	Description
APN	Access Point Name
BDS	BeiDou Navigation Satellite System
CEP	Circular Error Probable
CNR	Carrier-to-Noise Ratio
DOP	Dilution of Precision
eDRX	Extended Discontinuous Reception
EGPRS	Enhanced General Packet Radio Service
EPE	Estimated Position Error
EUFS	Extended User File Storage
FAQ	Frequently Asked Questions
Galileo	Galileo Satellite Navigation System (EU)
GGA	Global Positioning System Fix Data
GLONASS	Global Navigation Satellite System (Russian)

GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GSA	GNSS DOP and Active Satellites
GSM	Global System for Mobile Communications
GSV	GNSS Satellites in View
HDOP	Horizontal Dilution of Precision
IPv4	Internet Protocol Version 4
IPv6	Internet Protocol Version 6
LLE	Long Lasting Ephemeris
LPWA	Low-Power Wide Area
LTE	Long Term Evolution
LTE-M	LTE Machine-to-Machine Communications
MCC	Mobile Country Code
MCU	Microcontroller Unit/Microprogrammed Control Unit
ME	Mobile Equipment
MO	Mobile Originated
NB-IoT	Narrowband Internet of Things
NITZ	Network Identity and Time Zone
NMEA	NMEA (National Marine Electronics Association) 0183 Interface Standard
NTP	Network Time Protocol
NVRAM	Non-Volatile Random-Access Memory
PS	Packet Switch
PSM	Power Saving Mode
PTW	Paging Time Window
QZSS	Quasi-Zenith Satellite System

RAT	Radio Access Technology
RMC	Recommended Minimum Specific GNSS Data
RRC	Radio Resource Control
RSRP	Reference Signal Received Power
RSSI	Received Signal Strength Indicator
Rx	Receive
TCP	Transmission Control Protocol
TDM	Time-Division Multiplexing
TTFF	Time to First Fix
UART	Universal Asynchronous Receiver/Transmitter
UE	User Equipment
UFS	User File Storage
URL	Uniform Resource Locator
USB	Universal Serial Bus
UTC	Coordinated Universal Time
VTG	Course Over Ground & Ground Speed
Wi-Fi	Wireless Fidelity
WWAN	Wireless Wide Area Network
XOR	Exclusive-OR (gate)
XTRA	eXTended Receiver Assistance